

Spring REST using Spring Boot 3

1. spring-rest-handson

Create a Spring Web Project using Maven.

The screenshot shows the Spring Initializr web application in a browser. The URL is `start.spring.io`. The page has a dark theme and a sidebar with navigation icons. The main content area is divided into sections for Project, Language, Spring Boot, Project Metadata, Dependencies, and Configuration.

Project

- ☐ Gradle - Groovy
- ☐ Gradle - Kotlin
- ☒ Maven

Language

- ☒ Java
- ☐ Kotlin
- ☐ Groovy

Spring Boot

- ☐ 4.1.1 (SNAPSHOT)
- ☒ 4.1.0
- ☐ 4.0.8 (SNAPSHOT)
- ☐ 4.0.7
- ☒ 3.5.16

Project Metadata

Group: `com.cognizant`

Artifact: `spring_learn`

Package name: `com.cognizant.spring_learn`

Packaging: ☒ Jar ☐ War

Configuration: ☒ Properties ☐ YAML

Dependencies

Spring Web **WEB**

Build web, including RESTful, applications using Spring MVC. Uses Apache Tomcat as the default embedded container.

Spring Boot DevTools **DEVELOPER TOOLS**

Provides fast application restarts, LiveReload, and configurations for enhanced development experience.

Buttons: GENERATE (CTRL + G), EXPLORE (CTRL + SPACE), ...

The screenshot shows an IDE with the `SpringLearnApplication.java` file open. The file content is as follows:

```
1 package com.cognizant.spring_learn;
2
3 import org.springframework.boot.SpringApplication;
4
5 @SpringBootApplication
6 public class SpringLearnApplication {
7     public static void main(String[] args) { SpringApplication.run(SpringLearnApplication.class, args); }
8 }
9
10
```

The Run console shows the following output:

```
2026-07-13T12:15:49.852+05:30 INFO 13104 --- [spring_learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-13T12:15:49.852+05:30 INFO 13104 --- [spring_learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.55]
2026-07-13T12:15:49.894+05:30 INFO 13104 --- [spring_learn] [ restartedMain] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-07-13T12:15:49.895+05:30 INFO 13104 --- [spring_learn] [ restartedMain] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed
2026-07-13T12:15:50.330+05:30 INFO 13104 --- [spring_learn] [ restartedMain] o.s.b.d.a.OptionalLiveReloadServer : LiveReload server is running on port 35729
2026-07-13T12:15:50.375+05:30 INFO 13104 --- [spring_learn] [ restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2026-07-13T12:15:50.386+05:30 INFO 13104 --- [spring_learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Started SpringLearnApplication in 2.576 seconds (process...
```

SpringLearnApplication.java

```
1 package com.cognizant.spring_learn.controller;
2
3 import org.springframework.web.bind.annotation.GetMapping;
4 import org.springframework.web.bind.annotation.RestController;
5
6 @RestController
7 public class HelloController {
8
9     @GetMapping("/")
10    public String home() {
11        return "Welcome to Spring Boot Week 5";
12    }
13
14    @GetMapping("/hello")
15    public String hello() {
16        return "Hello Cognizant Digital Nurture";
17    }
18 }
```

Run SpringLearnApplication

```
2026-07-13T12:25:02.500+05:30 INFO 27980 --- [spring_learn] [ restartedMain] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed
2026-07-13T12:25:02.838+05:30 INFO 27980 --- [spring_learn] [ restartedMain] o.s.b.d.a.OptionalLiveReloadServer : LiveReload server is running on port 35729
2026-07-13T12:25:02.866+05:30 INFO 27980 --- [spring_learn] [ restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2026-07-13T12:25:02.875+05:30 INFO 27980 --- [spring_learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Started SpringLearnApplication in 2.01 seconds (process
2026-07-13T12:25:13.371+05:30 INFO 27980 --- [spring_learn] [nio-8080-exec-1] o.a.c.c.E.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-13T12:25:13.374+05:30 INFO 27980 --- [spring_learn] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 1 ms
```

spring_learn

```
1 <?xml version='1.0' encoding='UTF-8'?>
2 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3 <modelVersion>4.0.0</modelVersion>
4 <groupId>com.cognizant</groupId>
5 <artifactId>spring_learn</artifactId>
6 <version>0.0.1</version>
7 <packaging>war</packaging>
8 <name>spring_learn</name>
9 <description>Spring Boot Week 5</description>
10 <dependencies>
11 <dependency>
12 <groupId>org.springframework.boot</groupId>
13 <artifactId>spring-boot-devtools</artifactId>
14 <optional>true</optional>
15 </dependency>
16 <dependency>
17 <groupId>org.springframework.boot</groupId>
18 <artifactId>spring-boot-devtools</artifactId>
19 <scope>runtime</scope>
20 <optional>true</optional>
21 </dependency>
22 <dependency>
23 <groupId>org.springframework.boot</groupId>
24 <artifactId>spring-boot-starter-test</artifactId>
25 <scope>test</scope>
26 </dependency>
27 </dependencies>
28 <build>
29 <plugins>
30 <plugin>
31 <groupId>org.springframework.boot</groupId>
32 <artifactId>spring-boot-maven-plugin</artifactId>
33 </plugin>
34 </plugins>
35 </build>
36 </project>
```

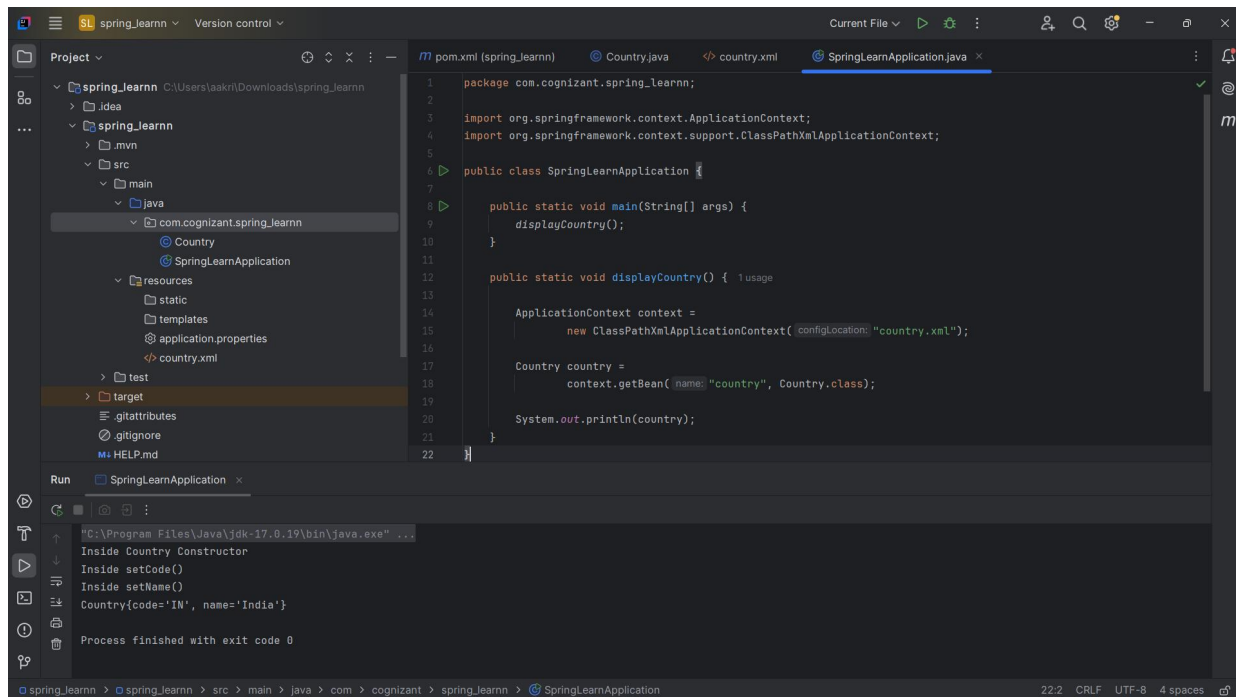
Run SpringLearnApplication

spring_learn

localhost:8080

Welcome to Spring Boot Week 5

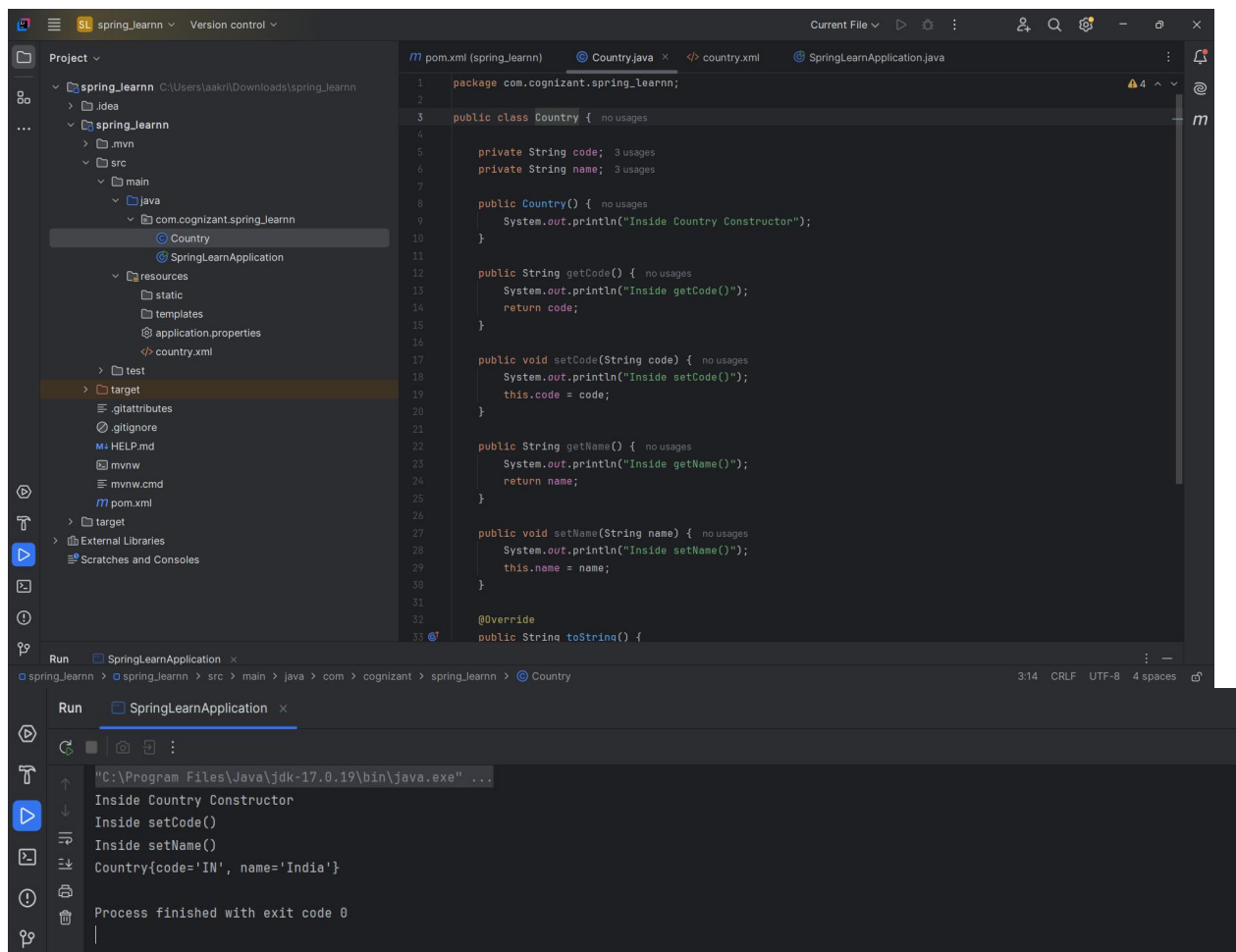
Spring Core – Load Country from Spring Configuration XML



```
1 package com.cognizant.spring_learnn;
2
3 import org.springframework.context.ApplicationContext;
4 import org.springframework.context.support.ClassPathXmlApplicationContext;
5
6 public class SpringLearnApplication {
7
8     public static void main(String[] args) {
9         displayCountry();
10    }
11
12    public static void displayCountry() {
13        ApplicationContext context =
14            new ClassPathXmlApplicationContext("country.xml");
15
16        Country country =
17            context.getBean("country", Country.class);
18
19        System.out.println(country);
20    }
21 }
22
```

Run: SpringLearnApplication

Process finished with exit code 0



```
1 package com.cognizant.spring_learnn;
2
3 public class Country {
4
5     private String code;
6     private String name;
7
8     public Country() {
9         System.out.println("Inside Country Constructor");
10    }
11
12    public String getCode() {
13        System.out.println("Inside getCode()");
14        return code;
15    }
16
17    public void setCode(String code) {
18        System.out.println("Inside setCode()");
19        this.code = code;
20    }
21
22    public String getName() {
23        System.out.println("Inside getName()");
24        return name;
25    }
26
27    public void setName(String name) {
28        System.out.println("Inside setName()");
29        this.name = name;
30    }
31
32    @Override
33    public String toString() {
34
35    }
36 }
37
```

Run: SpringLearnApplication

Process finished with exit code 0

